

PrintSmart

Unified Print Estimation & Inventory
System

PHASE 1 & 2 COMPREHENSIVE PRESENTATION

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DEFINING THE PROBLEM

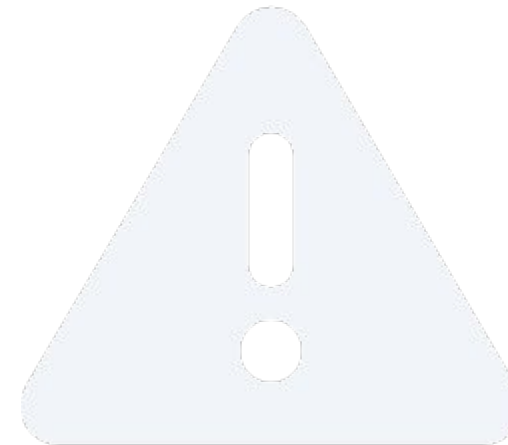
The current printing shop workflow is hindered by **manual dependency** and information gaps.

Bottleneck Operations

Employees rely on fixed Excel sheets that fail for custom orders, forcing constant consultation with the manager.

Information Disparity

Employees quote based on "worst-case" scenarios, while the owner holds exclusive metrics (wattage, material costs, B2B tiers).



IDENTIFIED ISSUES VS. PROJECT OBJECTIVES

Identified Critical Issues	System Strategic Objectives
Static pricing fails for dynamic requests.	Automate all dynamic custom pricing cases.
Managerial approval delays operations.	Enable full autonomy for front-desk staff.
Inability to calculate B2B loyalty rates.	Implement tiered B2B pricing logic.
No system for accurate inventory tracking.	Real-time tracking of material and stock.
Price inconsistency across visits.	Ensure uniform pricing via algorithm.

DEFINING THE SCOPE

IN-SCOPE

- Dynamic Price Estimation Engine
- Waste, Bleed, and Margin Calculations
- Inventory Tracking & Threshold Alerts
- Tiered Pricing for B2B/B2C Segments
- Role-Based Access Control (RBAC)

OUT-OF-SCOPE

- External Accounting Systems Integration
- Marketing and Sales Outreach Modules
- Human Resources (HR) Management
- Complex CRM Lead Generation
- Mobile App (Web-Based only for BYOD)

STAKEHOLDER ANALYSIS



Front Desk

Primary user handling sales and quick quotations.



Graphic Designer

Format designs and prepares print-ready files.



Machine Operator

Monitors production and maintenance logs.



Supervisor

Coordinates team and resolves edge cases.



Manager / Owner

Full admin privileges. Controls sensitive financial metrics, actual material costs, and B2B deal structures.

SYSTEM REQUIREMENTS

</> Functional

- **Dynamic Calculation:** Price based on material, waste, and bleed.
- **Inventory Deduction:** Auto-deduct stock upon order confirmation.
- **Data Concealment:** Prohibit non-admins from viewing actual costs.
- **Quotation Output:** Generate dated PDF quotations.

🔗 Non-Functional

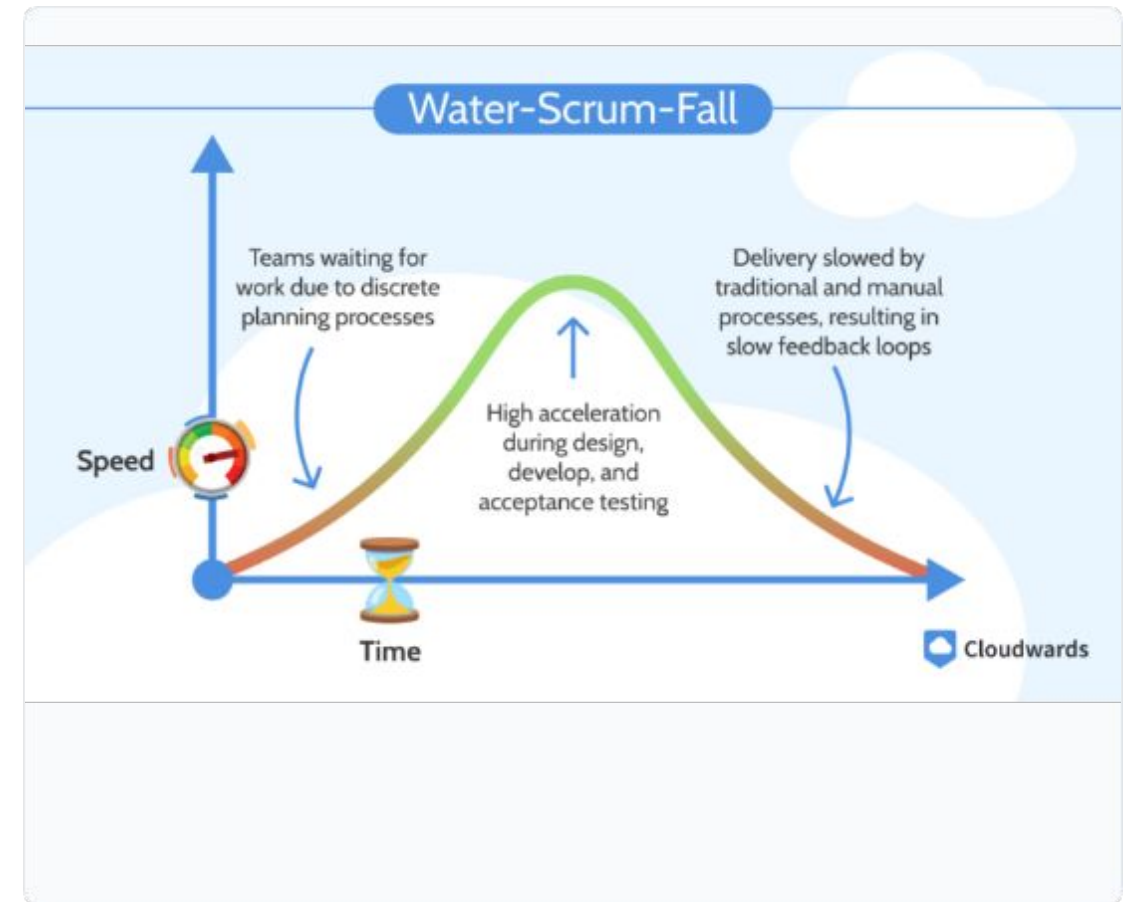
- **Usability:** Employee proficient within 2 hours of training.
- **Performance:** Results returned in < 3 seconds.
- **Accessibility:** Web-based to support BYOD.
- **Security:** RBAC for sensitive operational data.

METHODOLOGY: WATER-SCRUM-FALL

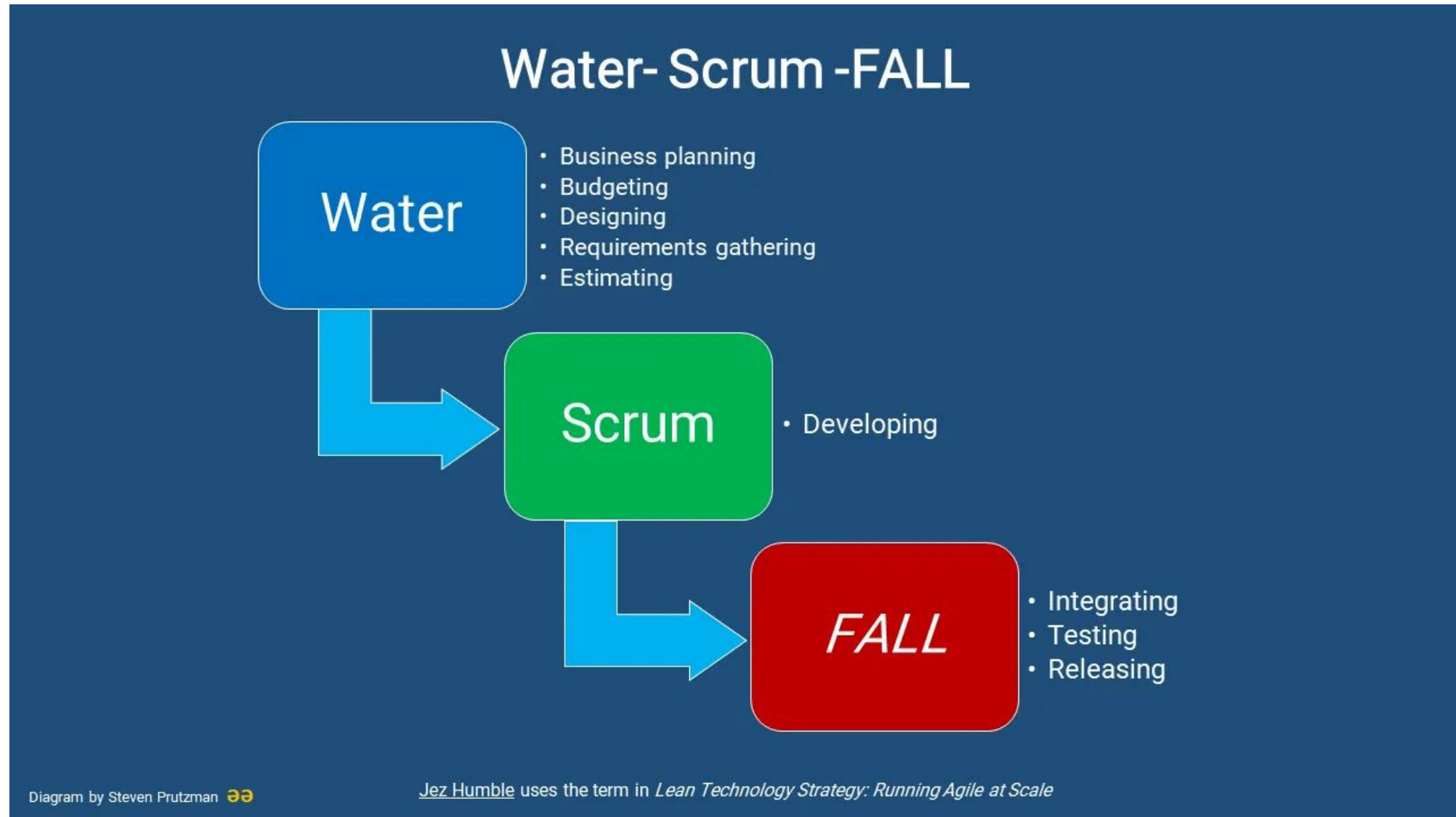
We adopted a **Hybrid Approach** where project management is driven by SDLC (Waterfall) while engineering uses iterative cycles (Scrum).

- **Water:** Upfront planning, budgeting, and architecture.
- **Scrum:** Sprints for rapid development of pricing modules.
- **Fall:** Gated releases, UAT, and production deployment.

Source: Forrester Research & "Is your business stuck in the Water-Scrum-Fall model?" (Medium).



METHODOLOGY: WATER-SCRUM-FALL



FUNCTION POINT ESTIMATION (UFP)

Component Type	Description	Complexity	Points
Inputs	Order details, Customer details, Stock input	Complex/Avg	18
Outputs	Quotations, Stock alerts, Waste reports	Complex/Avg	20
Files (Tables)	Stock, Machines, Customers, Orders	Mix	18
Inquiries	Search filtering for stock and orders	Average	20
Total Unadjusted Function Points (UFP)		76	

Adjusted FP (AFP) = $76 \times [0.65 + (35 \times 0.01)] = 76$ Adjusted FP

Estimate: $76 \text{ FP} / 18 \text{ productivity rate} = 4.2 \text{ Man-Months Effort}$

CRITIQUE OF FUNCTION POINT ANALYSIS

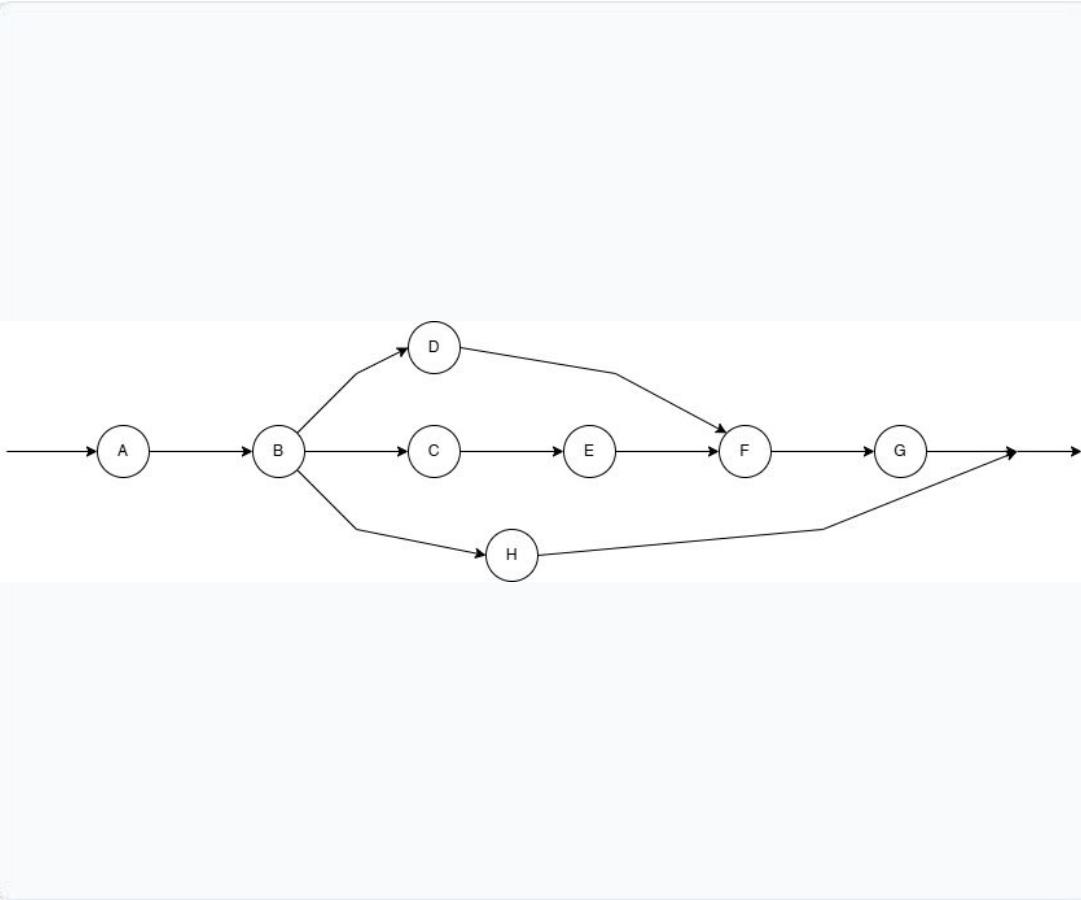
While standard, FPA is often **inaccurate in modern business** contexts like PrintSmart.

- **Ignores Non-Functional Complexity:** FPA misses effort for Security, Performance, and UX.
- **Linear Illusion:** Assumes complexity grows linearly with function count.
- **Technology Neutrality:** Fails to account for modern API/UI structural coupling.

"Function points fail to predict risk driven by structural coupling rather than functional volume."

— Software Industry Analysts (IN-COM)

PHASE 2: PERT & CRITICAL PATH



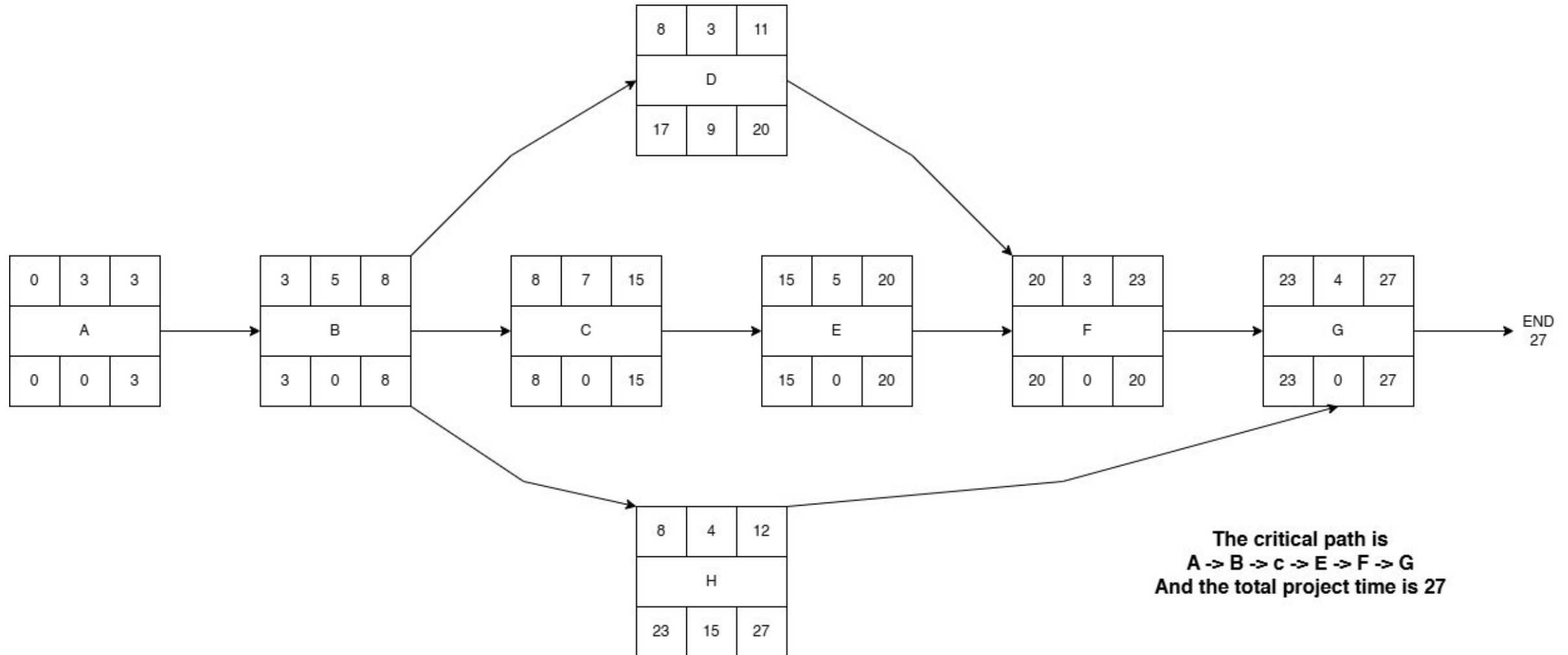
We mapped task dependencies to identify the minimum time required for completion.

Critical Path: A → B → C → E → F → G

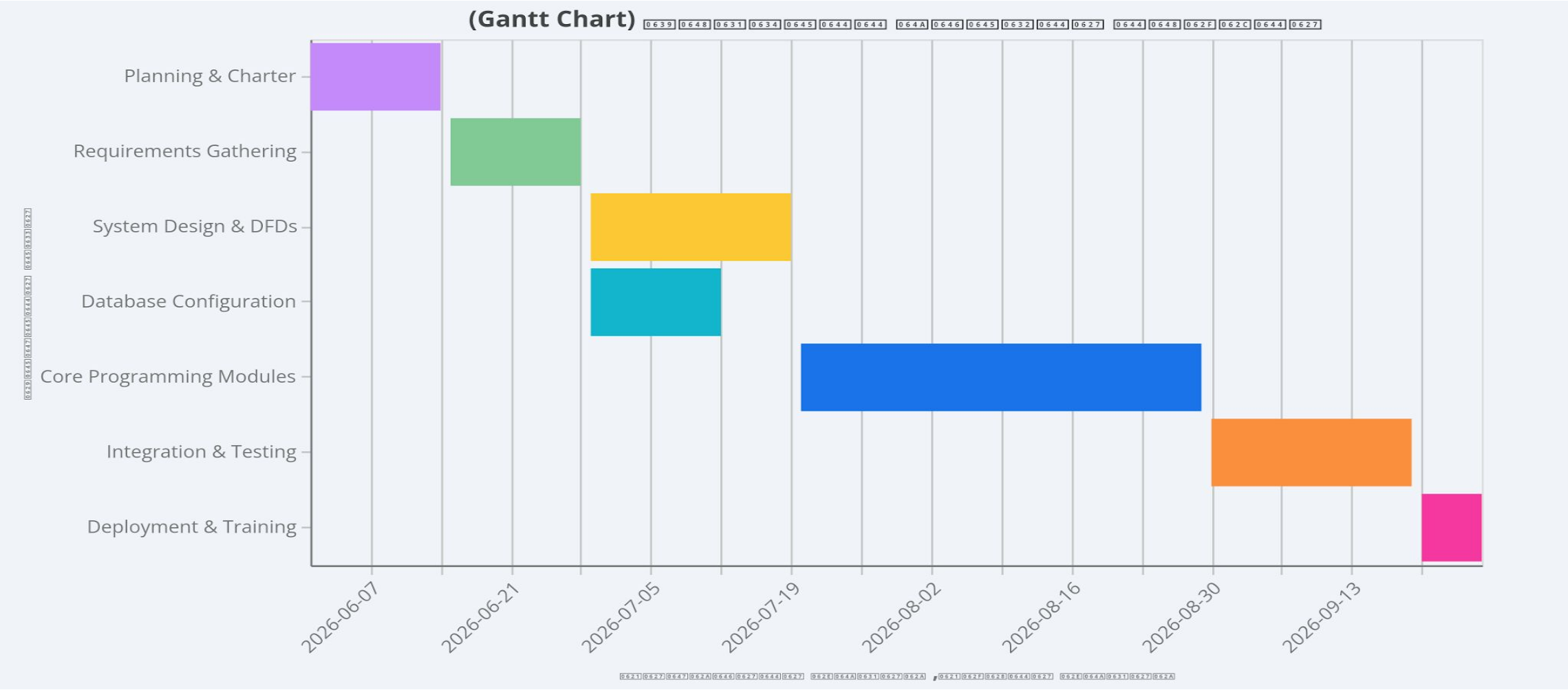
Total Duration: 27 Weeks

Tasks D and H possess slack time, allowing resource reallocation if needed.

PHASE 2: PERT & CRITICAL PATH



PHASE 2: PROJECT SCHEDULE (GANTT)



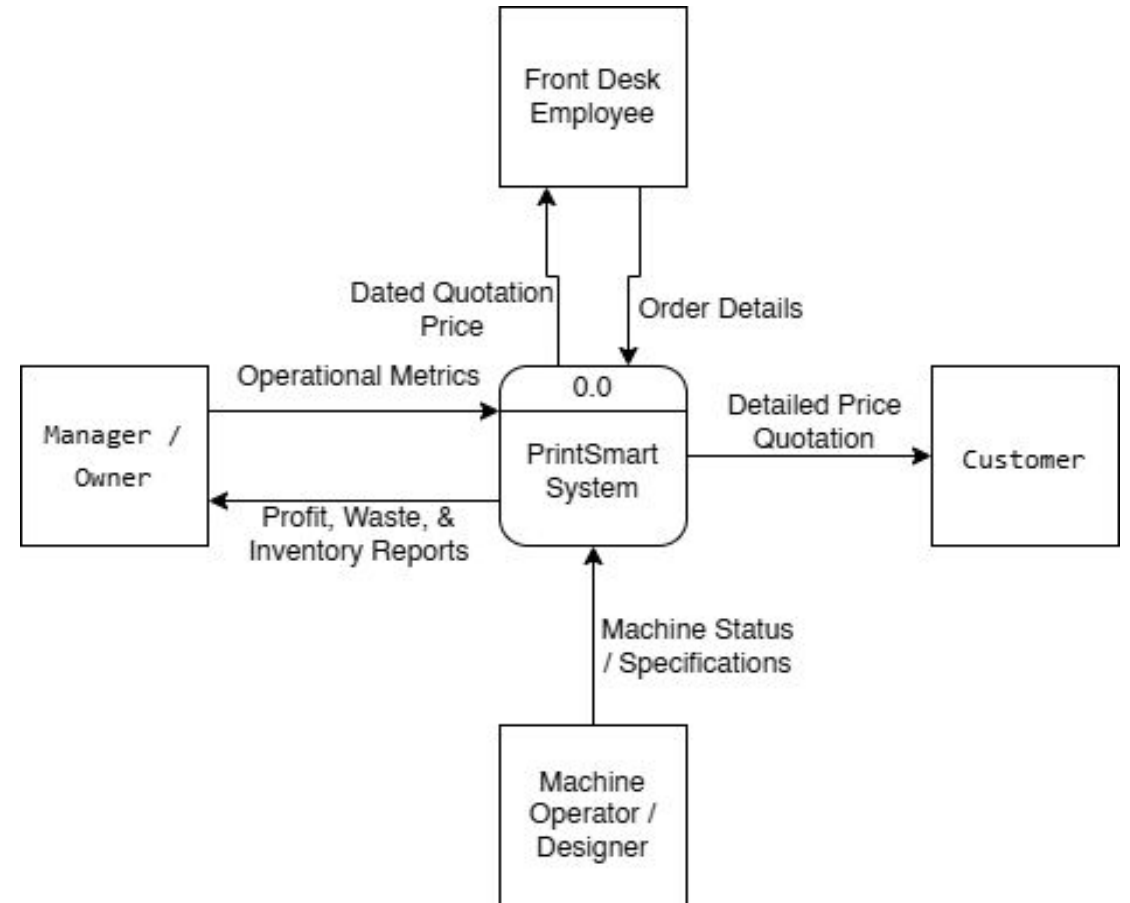
Visualizes the timeline from 12th June to late September, managing parallel task execution.

CONTEXT & LOGICAL DFD

Business Flow

Modeling *what* the system does from a business perspective.

- **Process 1.0:** Dynamic Pricing Logic.
- **Process 3.0:** Inventory Automation.
- **External Entities:** Unified interface for all roles.

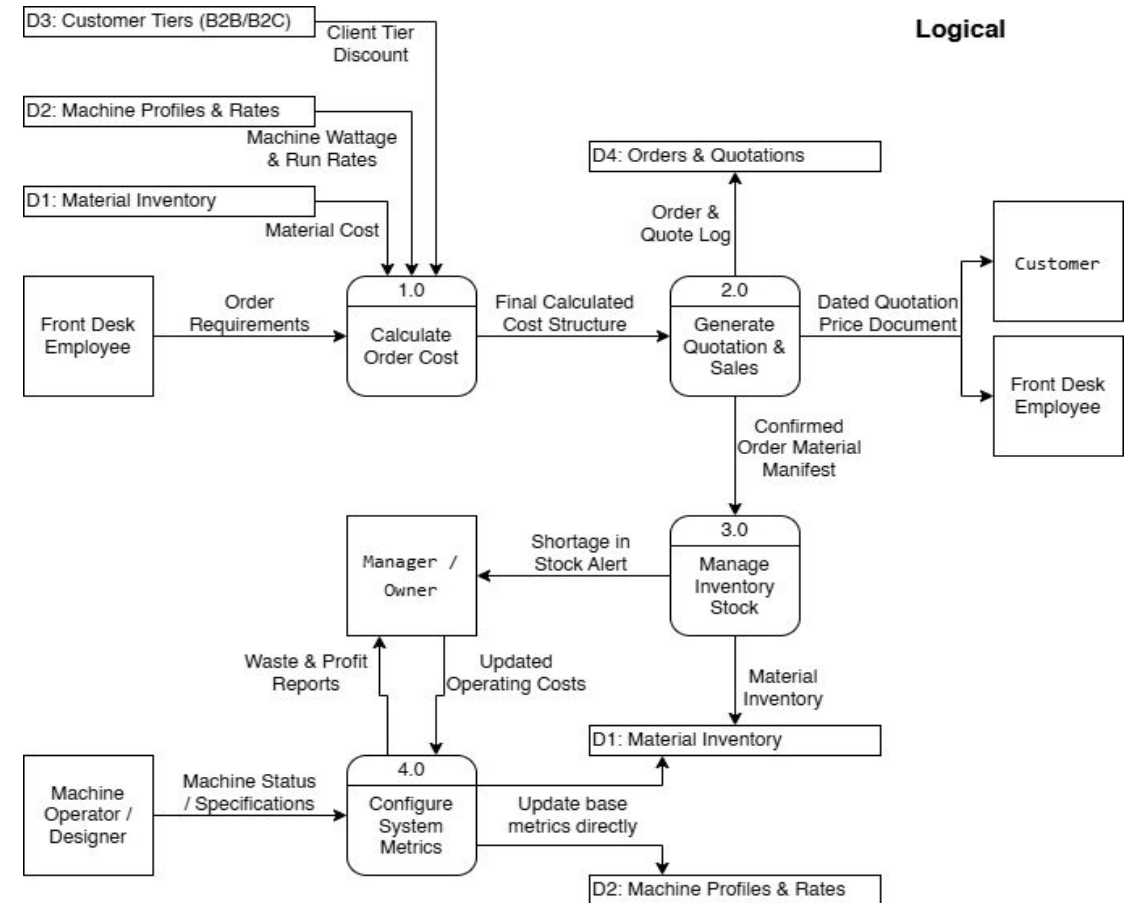


CONTEXT & LOGICAL DFD

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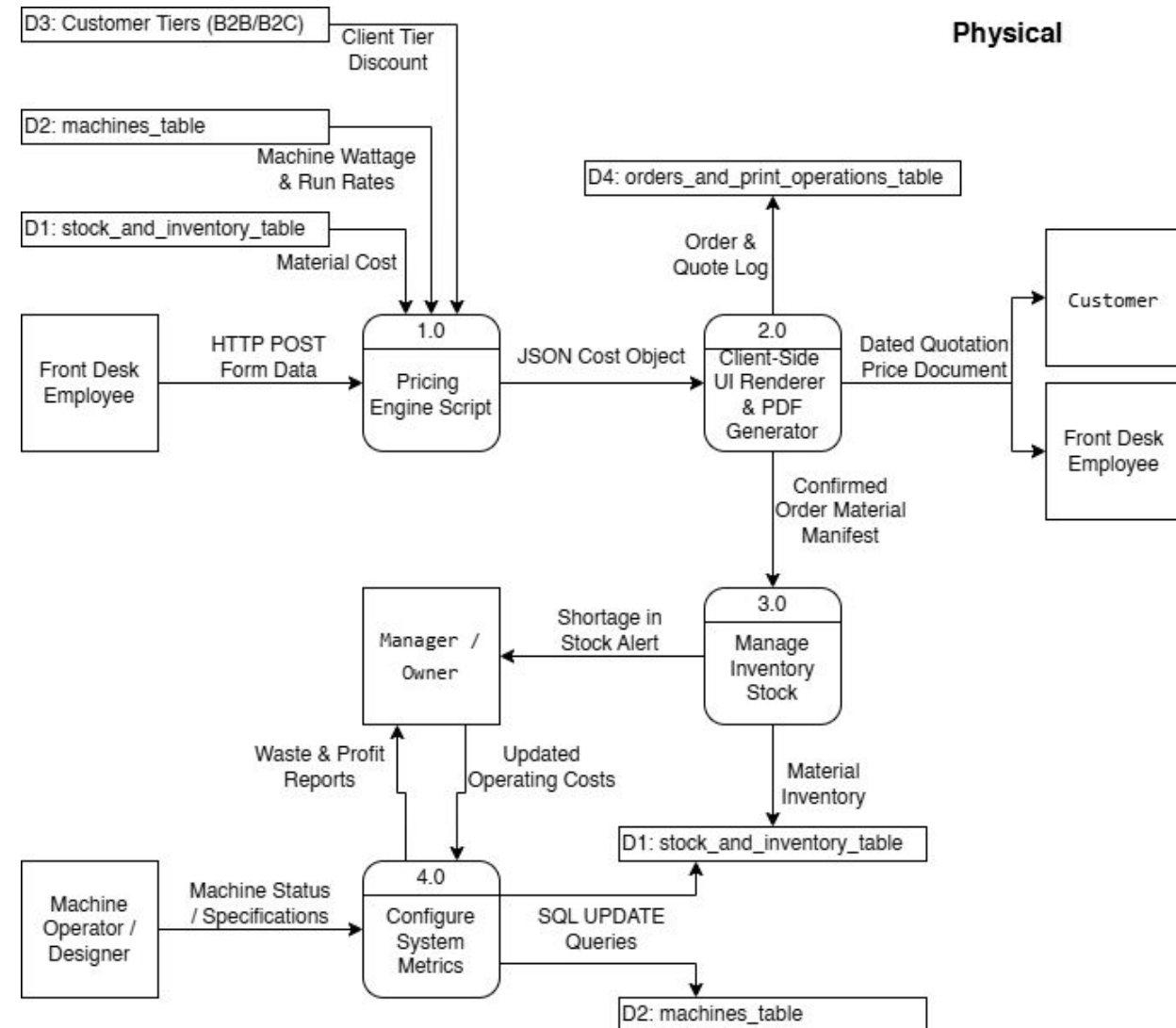
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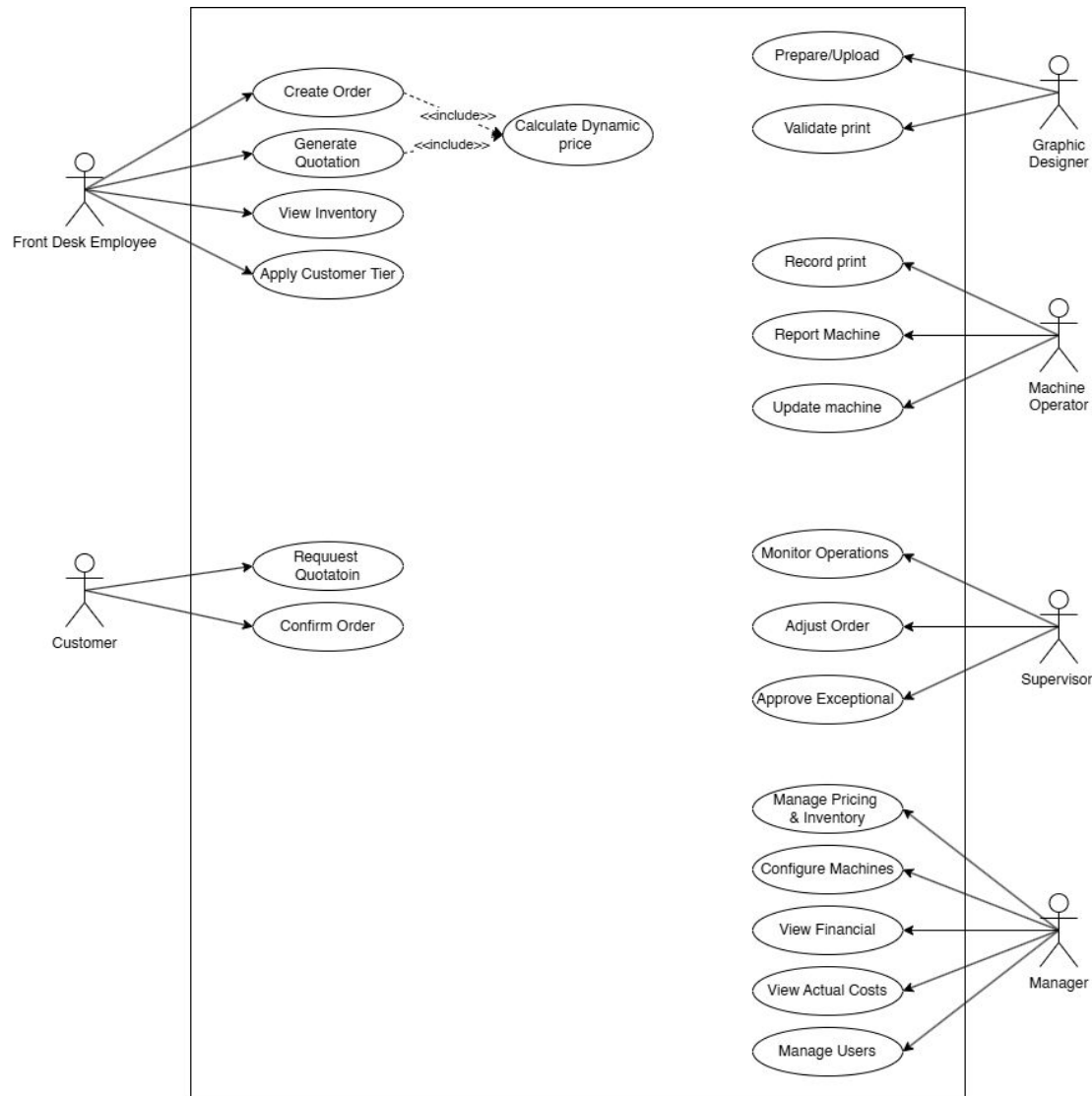
PHYSICAL DFD ARCHITECTURE

From Logic to Implementation

- HTTP POST Form Data for inputs.
- JSON Objects for dynamic pricing payloads.
- SQL UPDATE Queries for real-time inventory management.
- Transitioning logical stores to **Physical Database Tables**.



USE CASE: BEHAVIORAL MODELING



Bridges Structured Analysis with **Object-Oriented Design**.

- **Includes Relationships:** Linking Order Creation to Pricing Calculation.
- **Actor-Centric:** Visualizing boundaries for Front Desk, Designer, and Manager.
- **RBAC Visualization:** Ensuring "View Actual Costs" is exclusive to the Manager.

OPERATIONAL BEHAVIORAL LOGIC

Event	Source	Activity	Response
Request Quote	Front Desk	Calculate Dynamic Price	Dated PDF Quotation
Confirm Order	Front Desk	Create Order & Receipt	Job Ticket for Operator
Finish Job	Operator	Record Print / Update Stock	Production Waste Report
Low Stock Alert	Temporal	Inventory Auto-scan	Manager Alert
Financial Request	Manager	View Actual Costs	Secure Metrics Report

Thank You

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My Portfolio

